

$$\begin{aligned}\nabla^2 |f(z)|^2 &= 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} [f(z) \overline{f(z)}] \\ &= 4 \left[\frac{\partial}{\partial z} f(z) \right] \left[\frac{\partial}{\partial \bar{z}} \overline{f(z)} \right]\end{aligned}$$

[$\because f(z)$ is independent of $\bar{z}$ and $\overline{f(z)}$ is independent of z]

$$\begin{aligned}\therefore \nabla^2 |f(z)|^2 &= 4 [f'(z) \left[\frac{\partial}{\partial \bar{z}} \overline{f(z)} \right]] = 4 f'(z) \overline{f'(z)} \\ &= 4 |f'(z)|^2 \quad [\because z\bar{z} = |z|^2]\end{aligned}$$

Example: 3.20 Give an example such that u and v are harmonic but $u + iv$ is not analytic.

Solution:

$$u = x^2 - y^2, \quad v = \frac{-y}{x^2 + y^2}$$

Example: 3.21 Find the value of m if $u = 2x^2 - my^2 + 3x$ is harmonic.

Solution:

$$\text{Given } u = 2x^2 - my^2 + 3x$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad [\because u \text{ is harmonic}] \quad \dots (1)$$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= 4x + 3 \\ \frac{\partial^2 u}{\partial x^2} &= 4 \end{aligned} \right| \begin{aligned} \frac{\partial u}{\partial y} &= -2my \\ \frac{\partial^2 u}{\partial y^2} &= -2m \end{aligned}$$

$$\therefore (1) \Rightarrow (4) + (-2m) = 0$$

$$\Rightarrow m = 2$$

3.4 CONSTRUCTION OF ANALYTIC FUNCTION

There are three methods to find $f(z)$.

Method: 1 Exact differential method.

- (i) Suppose the harmonic function $u(x, y)$ is given.

Now, $dv = v_x dx + v_y dy$ is an exact differential

Where, v_x and v_y are known from u by using C-R equations.

$$\therefore v = \int v_x dx + \int v_y dy = - \int u_y dx + \int u_x dy$$

- (ii) Suppose the harmonic function $v(x, y)$ is given.

Now, $du = u_x dx + u_y dy$ is an exact differential

Where, u_x and u_y are known from v by using C-R equations.

$$\begin{aligned} u &= \int u_x dx + \int u_y dy \\ &= \int v_y dx + \int -v_x dy \\ &= \int v_y dx - \int v_x dy \end{aligned}$$

Method: 2 Substitution method

$$f(z) = 2u \left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a) \right] - [u(a, 0), -iv(a, 0)]$$

Here, $u(a, 0), -iv(a, 0)$ is a constant

$$\text{Thus } f(z) = 2u \left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a) \right] + C$$

By taking $a = 0$, that is, if $f(z)$ is analytic $z = 0 + i0$,

We have the simpler formula for $f(z)$

$$f(z) = 2 \left[u \frac{z}{2}, \frac{-iz}{2} \right] + C$$

Method: 3 [Milne – Thomson method]

(i) To find $f(z)$ when u is given

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= u_x - iv_y \text{ [by C-R condition]}$$

$$\therefore f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \text{ [by Milne-Thomson rule],}$$

Where, C is a complex constant.

(ii) To find $f(z)$ when v is given

$$\text{Let } f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$= v_y + iv_x \text{ [by C-R condition]}$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C \text{ [by Milne-Thomson rule],}$$

Where, C is a complex constant.

Example: 3.22 Construct the analytic function $f(z)$ for which the real part is $e^x \cos y$.

Solution:

$$\text{Given } u = e^x \cos y$$

$$\Rightarrow u_x = e^x \cos y \quad [\because \cos 0 = 1]$$

$$\Rightarrow u_x(z, 0) = e^x$$

$$\Rightarrow u_y = e^x \cos y \quad [\because \sin 0 = 0]$$

$$\Rightarrow u_y(z, 0) = 0$$

$$\therefore f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \text{ [by Milne-Thomson rule],}$$

Where, C is a complex constant.

$$\begin{aligned} \therefore f(z) &= \int e^z dz - i \int 0 dz + C \\ &= e^z + C \end{aligned}$$

Example: 3.23 Determine the analytic function $w = u + iv$ if $u = e^{2x}(x \cos 2y - y \sin 2y)$

Solution:

$$\text{Given } u = e^{2x}(x \cos 2y - y \sin 2y)$$

$$u_x = e^{2x}[\cos 2y] + (x \cos 2y - y \sin 2y)[2e^{2x}]$$

$$\begin{aligned}
 u_x(z, 0) &= e^{2z}[1] + [z(1) - 0][2e^{2z}] \\
 &= e^{2z} + 2ze^{2z} \\
 &= (1 + 2z)e^{2z}
 \end{aligned}$$

$$u_y = e^{2x}[-2x \sin 2y - (y2\cos 2y + \sin 2y)]$$

$$u_y(z, 0) = e^{2z}[-0 - (0 + 0)] = 0$$

$$\therefore f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$\begin{aligned}
 f(z) &= \int (1 + 2z)e^{2z}dz - i \int 0 + dz + C \\
 &= \int (1 + 2z)e^{2z}dz + C \\
 &= (1 + 2z)\frac{e^{2z}}{2} - 2\frac{e^{2z}}{4} + C \quad [\because \int uv dz = uv_1 - u'v_2 + u''v_3 - \dots] \\
 &= \frac{e^{2z}}{2} + ze^{2z} - \frac{e^{2z}}{2} + C \\
 &= ze^{2z} + C
 \end{aligned}$$

Example: 3.24 Determine the analytic function where real part is

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1.$$

Solution:

$$\text{Given } u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = 3x^2 - 3y^2 + 6x$$

$$\Rightarrow u_x(z, 0) = 3z^2 - 0 + 6z$$

$$u_y = 0 - 6xy + 0 - 6y$$

$$\Rightarrow u_y(z, 0) = 0$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$\begin{aligned}
 f(z) &= \int (3z^2 + 6z)dz - i \int 0 + dz + C \\
 &= 3\frac{z^2}{3} + 6\frac{z^2}{2} + C \\
 &= z^3 + 3z^2 + C
 \end{aligned}$$

Example: 3.25 Determine the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y - \cos 2x}$

Solution:

$$\text{Given } u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$u_x = \frac{(\cosh 2y - \cos 2x)[2 \cos 2x] - \sin 2x[2 \sin 2x]}{[\cosh 2y - \cos 2x]^2}$$

$$u_x(z, 0) = \frac{(1 - \cos 2z)(2 \cos 2z) - 2 \sin^2 2z}{[\cosh 0 - \cos 2z]^2}$$

$$= \frac{2 \cos 2z - 2 \cos^2 2z - 2 \sin^2 2z}{(1 - \cos 2z)^2}$$

$$\begin{aligned}
&= \frac{2 \cos 2z - 2[\cos^2 2z + \sin^2 2z]}{(1 - \cos 2z)^2} \\
&= \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2} \\
&= \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2} \\
&= \frac{2 \cos 2z - 2}{(1 - \cos 2z)} \\
&= \frac{-2}{2 \sin^2 z} \\
&= -\operatorname{cosec}^2 z
\end{aligned}$$

$$u_y = \frac{(\cosh 2y - \cos 2x)(0) - \sin 2x[2 \sin 2y]}{[\cosh 2y - \cos 2x]^2}$$

$$\Rightarrow u_y(z, 0) = 0$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}],$$

where C is a complex constant.

$$\begin{aligned}
f(z) &= \int (-\operatorname{cosec}^2 z) dz - i \int 0 dz + C \\
&= \cot z + C
\end{aligned}$$

Example: 3.26 Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic and determine its conjugate.

Also find $f(z)$

Solution:

$$\text{Given } u = \frac{1}{2} \log(x^2 + y^2)$$

$$u_x = \frac{1}{2} \frac{1}{(x^2 + y^2)} (2x) = \frac{x}{x^2 + y^2},$$

$$\Rightarrow u_x(z, 0) = \frac{z}{z^2} = \frac{1}{z}$$

$$u_{xx} = \frac{(x^2 + y^2)[1] - x[2x]}{[x^2 + y^2]^2} = \frac{x^2 + y^2 - 2x^2}{[x^2 + y^2]^2} = \frac{y^2 - x^2}{[x^2 + y^2]^2} \quad \dots (1)$$

$$u_y = \frac{1}{2} \frac{1}{x^2 + y^2} (2y) = \frac{y}{x^2 + y^2}$$

$$\Rightarrow u_y(z, 0) = 0$$

$$u_{yy} = \frac{(x^2 + y^2)[1] - y[2y]}{[x^2 + y^2]^2} = \frac{x^2 - y^2}{[x^2 + y^2]^2} \quad \dots (2)$$

To prove u is harmonic:

$$\therefore u_{xx} + u_{yy} = \frac{(y^2 - x^2) + (x^2 - y^2)}{[x^2 + y^2]^2} = 0 \quad \text{by (1) \& (2)}$$

$\Rightarrow u$ is harmonic.

To find $f(z)$:

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}],$$

Where, C is a complex constant.

$$f(z) = \int \frac{1}{z} dz - i \int 0 dz + C$$

To find v :

$$f(z) = \log(re^{i\theta}) \quad [\because z = re^{i\theta}]$$

$$u + iv = \log r + \log e^{i\theta} = \log r + i\theta$$

$$\Rightarrow u = \log r, v = \theta$$

Note: $z = x + iy$

$$r = |z| = \sqrt{x^2 + y^2}$$

$$\log r = \frac{1}{2} \log(x^2 + y^2)$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad i.e., v = \tan^{-1}\left(\frac{y}{x}\right)$$

Example: 3.27 Construct an analytic function $f(z) = u + iv$, given that

$u = e^{x^2-y^2} \cos 2xy$. Hence find v .

Solution:

$$\text{Given } u = e^{x^2-y^2} \cos 2xy = e^{x^2} e^{-y^2} \cos 2xy$$

$$u_x = e^{-y^2} [e^{x^2} (-2y \sin 2xy) + \cos 2xy e^{x^2} 2x]$$

$$u_x(z, 0) = 1 [e^{z^2} (0) + 2ze^{z^2}] = 2ze^{z^2}$$

$$u_y = e^{x^2} [e^{-y^2} (-2x \sin 2xy) + \cos 2xy e^{-y^2} (-2y)]$$

$$u_y(z, 0) = e^{z^2} [0 + 0] = 0$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

$$= \int 2z e^{z^2} dz + C$$

$$= 2 \int z e^{z^2} dz + C$$

$$\text{put } t = z^2, dt = 2z dz$$

$$= \int e^t dt + C$$

$$= e^t + C$$

$$f(z) = e^{z^2} + C$$

To find v :

$$u + iv = e^{(x+iy)^2} = e^{x^2-y^2+i2xy} = e^{x^2-y^2} e^{i2xy}$$

$$= e^{x^2-y^2} [\cos(2xy) + i \sin(2xy)]$$

$$v = e^{x^2-y^2} \sin 2xy \quad [\because \text{equating the imaginary parts}]$$

Example: 3.28 Find the regular function whose imaginary part is

$e^{-x}(x \cos y + y \sin y)$.

Solution:

$$\text{Given } v = e^{-x}(x \cos y + y \sin y)$$

$$v_x = e^{-x} [\cos y] + (x \cos y + y \sin y) [-e^{-x}]$$

$$v_x(z, 0) = e^{-z} + (z)(-e^{-z}) = (1 - z)e^{-z}$$

$$v_y = e^{-x}[-x \sin y + (y \cos y + \sin y (1))]$$

$$v_x(z, 0) = e^{-z}[0 + 0 + 0] = 0$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$\begin{aligned} f(z) &= \int 0dz + i \int (1 - z)e^{-z}dz + C \\ &= i \int (1 - z)e^{-z}dz + C \\ &= i \left[(1 - z) \left[\frac{e^{-z}}{-1} \right] - (-1) \left[\frac{e^{-z}}{(-1)^2} \right] \right] + C \\ &= i[-(1 - z)e^{-z} + e^{-z}] + C \\ &= iz e^{-z} + C \end{aligned}$$

Example: 3.29 In a two dimensional flow, the stream function is $\psi = \tan^{-1} \left(\frac{y}{x} \right)$. Find the velocity potential ϕ .

Solution:

$$\text{Given } \psi = \tan^{-1}(y/x)$$

We should denote, ϕ by u and ψ by v

$$\therefore v = \tan^{-1}(y/x)$$

$$v_x = \frac{1}{1+(y/x)^2} \left[\frac{-y}{x^2} \right] = \frac{-y}{x^2+y^2},$$

$$v_x(z, 0) = 0$$

$$v_y = \frac{1}{1+(y/x)^2} \left[\frac{1}{x} \right] = \frac{x}{x^2+y^2}$$

$$v_x(z, 0) = \frac{z}{z^2} = \frac{1}{z}$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C$$

$$f(z) = \int \frac{1}{z} dz + i \int 0 dz + C = \log z + C$$

To find ϕ :

$$f(z) = \log(re^{i\theta}) \quad [\because z = re^{i\theta}]$$

$$u + iv = \log r + \log e^{i\theta}$$

$$u + iv = \log r + i\theta$$

$$\Rightarrow u = \log r$$

$$\Rightarrow u = \log \sqrt{x^2 + y^2}$$

$$= \frac{1}{2} \log(x^2 + y^2)$$

$$z = x + iy, |z| = \sqrt{x^2 + y^2}$$

So, the velocity potential ϕ is

$$\phi = \frac{1}{2} \log(x^2 + y^2)$$

Note: In two dimensional steady state flows, the complex potential

$f(z) = \phi(x, y) + i\psi(x, y)$ is analytic.

Example: 3.30 If $w = u + iv$ is an analytic function and $v = x^2 - y^2 + \frac{x}{x^2+y^2}$, find u .

Solution:

$$\text{Given } v = x^2 - y^2 + \frac{x}{x^2+y^2}$$

$$v_x = 2x - 0 + \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2}$$

$$= 2x + \frac{y^2 - x^2}{(x^2+y^2)^2}, \quad v_x(z, 0) = 2z + \frac{(-z^2)}{(z^2)}$$

$$\Rightarrow v_x(z, 0) = 2z - \frac{1}{z^2}$$

$$v_y = 0 - 2y + \frac{0 - x(2y)}{(x^2+y^2)^2}$$

$$= 0 - 2y - \frac{2xy}{(x^2+y^2)^2}$$

$$\Rightarrow v_y(z, 0) = 0$$

$$\therefore f(z) = \int v_y(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int 0dz + i \int \left(2z - \frac{1}{z^2}\right) dz + C$$

$$= i \left[2 \frac{z^2}{2} + \frac{1}{z}\right] + C \quad \left[\because \int \frac{-1}{z^2} dz = \frac{1}{z}\right]$$

$$= i \left[z^2 + \frac{1}{z}\right] + C$$

Example: 3.31 If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .

Solution:

$$\text{Given } u - v = e^x(\cos y - \sin y), \quad \dots (A)$$

Differentiate (A) p.w.r. to x , we get

$$u_x - v_x = e^x(\cos y - \sin y),$$

$$u_x(z, 0) - v_x(z, 0) = e^z \quad \dots (1)$$

Differentiate (A) p.w.r. to y , we get

$$u_y - v_y = e^x(-\sin y - \cos y)$$

$$u_y(z, 0) - v_y(z, 0) = e^z[-1]$$

$$\text{i.e., } u_y(z, 0) - v_y(z, 0) = -e^z$$

$$-v_x(z, 0) - u_x(z, 0) = -e^z \quad \dots (2) \quad [\text{by C-R conditions}]$$

$$(1) + (2) \Rightarrow -2v_x(z, 0) = 0$$

$$\Rightarrow v_x(z, 0) = 0$$

$$(1) \Rightarrow u_x(z, 0) = e^z$$

$$f(z) = \int u_x(z, 0)dz + i \int v_x(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

$$f(z) = \int e^z dz + i0 + C$$

$$= e^z + C$$

Example: 3.32 Find the analytic functions $f(z) = u + iv$ given that

(i) $2u + v = e^x(\cos y - \sin y)$

(ii) $u - 2v = e^x(\cos y - \sin y)$

Solution:

Given (i) $2u + v = e^x(\cos y - \sin y) \quad \dots (A)$

Differentiate (A) p.w.r. to x, we get

$$2u_x + v_x = e^x(\cos y - \sin y)$$

$$2u_x - u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$2u_x(z, 0) - u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate (A) p.w.r. to y, we get

$$2u_y + v_y = e^x[-\sin y - \cos y]$$

$$2u_y + u_x = e^x[-\sin y - \cos y] \quad [\text{by C-R condition}]$$

$$2u_y(z, 0) + u_x(z, 0) = e^z(-1) = -e^z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 4u_x(z, 0) - 2u_y(z, 0) = 2e^z \quad \dots (3)$$

$$(2) + (3) \Rightarrow 5u_x(z, 0) = e^z$$

$$\Rightarrow u_x(z, 0) = \frac{1}{5}e^z$$

$$(1) \Rightarrow u_y(z, 0) = \frac{2}{5}e^z - e^z = -\frac{3}{5}e^z$$

$$\Rightarrow u_y(z, 0) = -\frac{3}{5}e^z$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int \frac{1}{5}e^z dz - i \int -\frac{3}{5}e^z dz + C$$

$$= \frac{2}{5}e^z + \frac{3}{5}ie^z + C$$

$$= \frac{1+3i}{5}e^z + C$$

(ii) $u - 2v = e^x(\cos y - \sin y) \quad \dots (B)$

Differentiate (B) p.w.r. to x, we get

$$u_x - 2v_x = e^x(\cos y - \sin y)$$

$$u_x + 2u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$u_x(z, 0) + 2u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate (B) p.w.r. to y, we get

$$u_y - 2v_y = e^x[-\sin y - \cos y]$$

$$u_y - 2u_x = e^x[-\sin y - \cos y] \quad [\text{by C-R condition}]$$

$$u_y(z, 0) - 2u_x(z, 0) = -e^z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 2u_x(z, 0) + 4u_y(z, 0) = 2e^z \quad \dots (3)$$

$$(2) + (3) \Rightarrow 5u_y(z, 0) = e^z$$

$$\Rightarrow u_y(z, 0) = \frac{1}{5} e^z$$

$$(1) \Rightarrow u_x(z, 0) = -\frac{2}{5} e^z + e^z$$

$$= \frac{3}{5} e^z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant.

$$f(z) = \int \frac{3}{5} e^z dz - i \int \frac{1}{5} e^z dz + C$$

$$= \frac{3}{5} e^z - i \frac{1}{5} e^z + C = \frac{3-i}{5} e^z + C$$

Example: 3.33 Determine the analytic function $f(z) = u + iv$ given that

$$3u + 2v = y^2 - x^2 + 16xy$$

Solution:

$$\text{Given } 3u + 2v = y^2 - x^2 + 16xy \quad \dots (A)$$

Differentiate (A) p.w.r. to x, we get

$$3u_x + 2v_x = -2x + 16y$$

$$3u_x - 2u_y = -2x + 16y \quad [\text{by C-R condition}]$$

$$3u_x(z, 0) - 2u_y(z, 0) = -2z \quad \dots (1)$$

Differentiate (A) p.w.r. to y, we get

$$3u_y + 2v_y = 2y + 16x$$

$$3u_y + 2u_x = 2y + 16x \quad [\text{by C-R condition}]$$

$$3u_y(z, 0) + 2u_x(z, 0) = 16z \quad \dots (2)$$

$$(1) \times (2) \Rightarrow 6u_x(z, 0) - 4u_y(z, 0) = -4z \quad \dots (3)$$

$$(2) \times (3) \Rightarrow 9u_y(z, 0) + 6u_x(z, 0) = 48z$$

$$(3) - (4) \Rightarrow -13u_y(z, 0) = -52z$$

$$\Rightarrow u_y(z, 0) = 4z$$

$$(1) \Rightarrow 3u_x(z, 0) = 8z - 2z = 6z$$

$$\Rightarrow u_x(z, 0) = 2z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

where C is a complex constant.

$$f(z) = \int 2z dz - i \int 4z dz + C$$

$$= 2 \frac{z^2}{2} - i \frac{4z^2}{2} + C$$

$$= z^2 - i2z^2 + C$$

$$= (1 - i2)z^2 + C$$

Example:3.34 Find an analytic function $f(z) = u + iv$ given that $2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$

Solution:

$$\text{Given } 2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

Differentiate p.w.r. to x , we get

$$2u_x + 3v_x = \frac{(\cosh 2y - \cos 2x)(2 \cos 2x) - \sin 2x (2 \sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$$2u_x - 3u_y = \frac{(\cosh 2y - \cos 2x)(2 \cos 2x) - \sin 2x (2 \sin 2x)}{(\cosh 2y - \cos 2x)^2} \quad [\text{by C-R condition}]$$

$$2u_x(z, 0) - 3u_y(z, 0) = \frac{2 \cos 2z(1 - \cos 2z) - 2 \sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2 \cos 2z - 2 \cos^2 2z - 2 \sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2 \cos 2z - 2}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z}$$

$$= \frac{-2}{2 \sin^2 z} = -\operatorname{cosec}^2 z$$

$$2u_x(z, 0) - 3u_y(z, 0) = -\operatorname{cosec}^2 z \quad \dots (1)$$

Differentiate p.w.r. to y , we get

$$2u_y + 3v_y = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2} \quad (2)$$

$$2u_y + 3u_x = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2} \quad [\text{by C - R condition}]$$

$$2u_y(z, 0) + 3u_x(z, 0) = 0 \quad \dots (2)$$

Solving (1) & (2) we get,

$$\Rightarrow u_x(z, 0) = -\frac{2}{13} \operatorname{cosec}^2 z$$

$$\Rightarrow u_y(z, 0) = -\frac{2}{13} \operatorname{cosec}^2 z$$

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + C \quad [\text{by Milne-Thomson rule}]$$

Where, C is a complex constant

$$f(z) = \int \left(\frac{-2}{13} \right) \operatorname{cosec}^2 z dz - i \int \left(\frac{3}{13} \right) \operatorname{cosec}^2 z dz + C$$

$$= \left(\frac{2}{13} \right) \cot z + \left(\frac{3}{13} \right) \cot z + C$$

$$= \frac{2+3i}{13} \cot z + C$$

Example: 3.35 Find the analytic function $f(z) = u + iv$ given that $2u + 3v = e^x(\cos y - \sin y)$

Solution:

$$\text{Given } 2u + 3v = e^x(\cos y - \sin y)$$

Differentiate p.w.r. to x , we get

$$2u_x + 3v_x = e^x(\cos y - \sin y)$$

$$2u_x - 3u_y = e^x(\cos y - \sin y) \quad [\text{by C-R condition}]$$

$$2u_x(z, 0) - 3u_y(z, 0) = e^z \quad \dots (1)$$

Differentiate p.w.r. to y, we get

$$2u_y + 3v_y = e^x[-\sin y - \cos y]$$

$$2u_y + 3u_x = -e^x [\sin y + \cos y] \quad [\text{by C-R condition}]$$

$$2u_y(z, 0) + 3u_x(z, 0) = -e^z \quad \dots (2)$$

$$(1) \times (3) \Rightarrow 6u_x(z, 0) - 9u_y(z, 0) = 3e^z \quad \dots (3)$$

$$(2) \times 2 \Rightarrow 6u_x(z, 0) + 4u_y(z, 0) = -2e^z \quad \dots (4)$$

$$(3) - (4) \Rightarrow -13u_y(z, 0) = 5e^z$$

$$\Rightarrow u_y(z, 0) = -\frac{5}{13}e^z$$

$$(1) \Rightarrow 2u_x(z, 0) + \frac{15}{13}e^z = e^z$$

$$2u_x(z, 0) = e^z - \frac{15}{13}e^z = -\frac{2}{13}e^z$$

$$\Rightarrow u_x(z, 0) = -\frac{1}{13}e^z$$

$$f(z) = \int u_x(z, 0)dz - i \int u_y(z, 0)dz + C$$

$$\therefore f(z) = \int \frac{-1}{13}e^z dz - i \int \left(\frac{-5}{13}\right) dz + C$$

$$= \frac{-1}{13}e^z + \frac{5}{13}e^z i + C = \frac{-1+5i}{13}e^z + C$$

Exercise: 3.4

Construction of an analytic function

1. Show that the function $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Find the conjugate harmonic function v and express $u + iv$ as an analytic function of z .

[Ans: $v(x, y) = 3x^2y + 4xy - x^3 + C$, $f(z) = -iz^3 + 2z^2 + iC$ where C is a real constant.]

2. If $f(z) = u + iv$ is an analytic function of z , and if $u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$, find v .

$$[\text{Ans: } v = \frac{-2 \sinh 2y}{e^{2y} + e^{-2y} - 2 \cos 2x} + C]$$

3. Find v such that $w = u + iv$ is an analytic function of z , given that $u = e^{x^2-y^2} \cos 2xy$. Hence find w .

$$[\text{Ans: } v = e^{x^2-y^2} \sin 2xy + C \quad w = e^{z^2} + C]$$

4. Find the analytic function $w = u + iv$ if $w = e^{2x}(x \cos 2y - y \sin 2y)$. Hence find u .

$$[\text{Ans: } w = iz e^{2z} + C, u = -(x \sin 2y + y \cos 2y) e^{2x} + C]$$

5. If $v = \frac{x-y}{x^2+y^2}$ find u such $u + iv$ is an analytic function. What is the harmonic conjugate of v ?

$$[\text{Ans: } u = \frac{x+y}{x^2+y^2} + C \quad \text{Harmonic conjugate of } v \text{ is } -u = \frac{-(x+y)}{x^2+y^2}, f(z) = \frac{1+i}{z} + C]$$

6. Find the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y + \cos 2x}$

$$[\text{Ans: } f(z) = \tan z + C]$$